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Lab 8: PIV

Overview

In Lab 8, we utilized a diode laser sheet, a vortex generator, and a camera to study the 2-dimensional velocity fields and the resulting movement of water particles due to a vortex. Videos were analyzed computationally with MATLAB's PIVLab software to visualize velocity fields and particle trajectories

Experimental Setup

- Air vortex gun
- Tripod
- Water particles
- Diode laser sheet
- Phone camera

The data processing used the following software:

- MATLAB
- MATLAB PIVLab

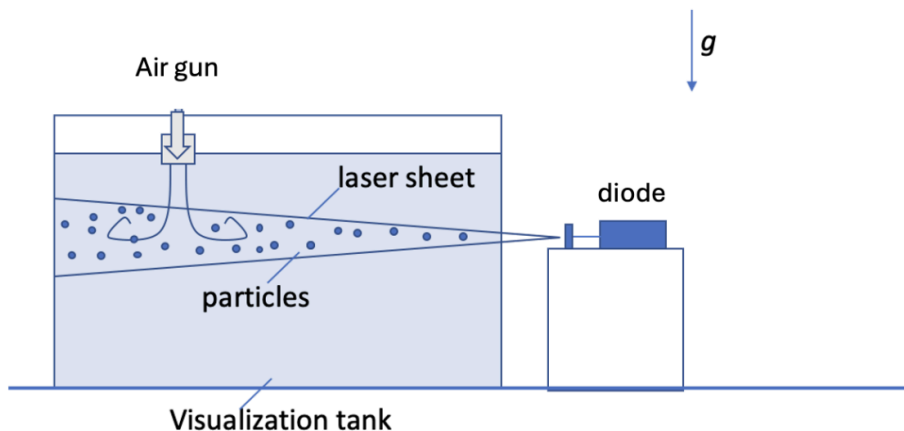


Figure 1 illustrates the particle image velocimetry (PIV) experiment. A vortex ring is generated by a air vortex gun inside a water-filled tank seeded with tracer particles. A diode laser projects a horizontal laser sheet to illuminate the particles. A phone camera, recording at 240 fps, captures the motion for later analysis in PIVLab

Data Analysis

Frame N at $t = 3.000$ s (red) and Frame N+1 at $t = 3.033$ s (green)

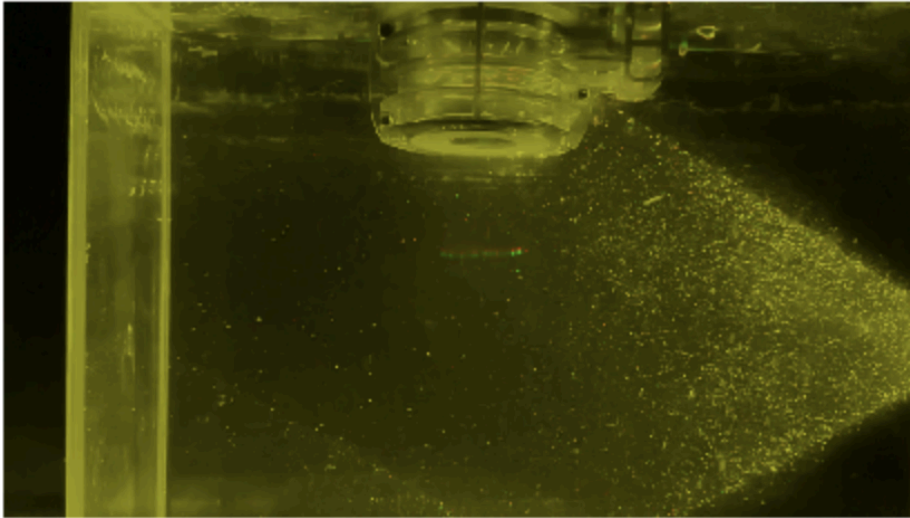


Figure 2 illustrates an overlay of two consecutive frames (130 and 131) showing that the vortex ring has barely shifted. The particles appear nearly identical between frames, indicating strong correlation and minimal motion, consistent with the early stage of the vortex's movement.

Frame N at $t = 3.000$ s (red) and Frame N+10 at $t = 3.333$ s (green)

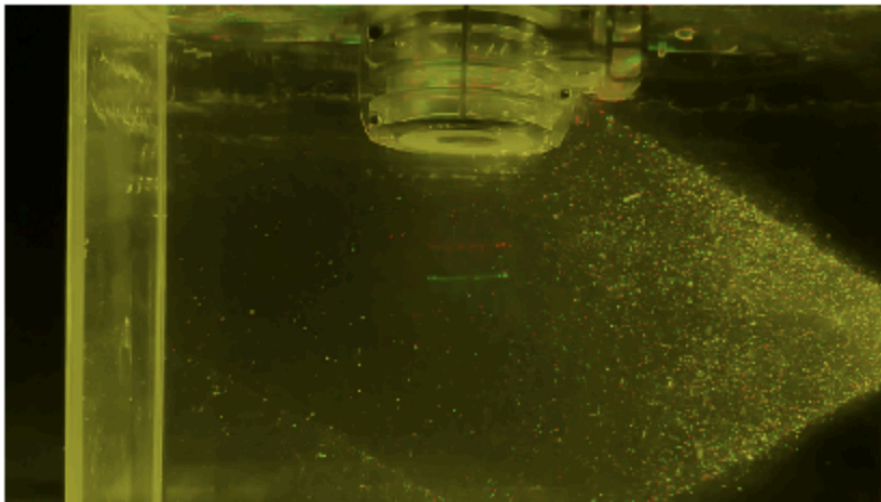
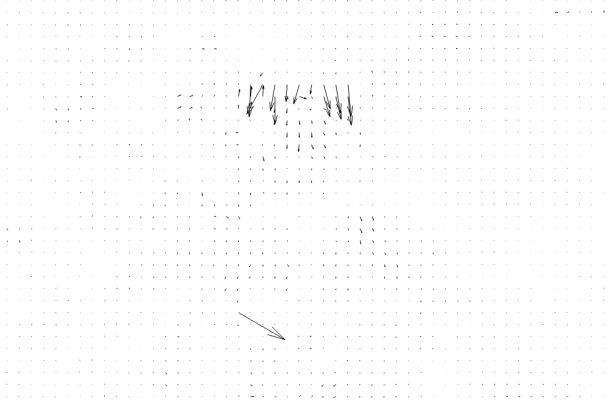
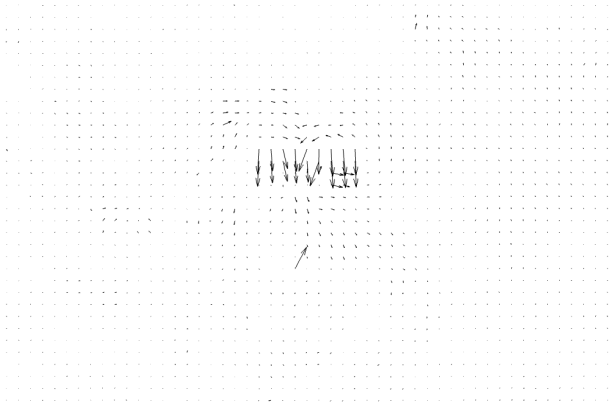


Figure 3 illustrates an overlay of frames 130 and 145 demonstrating a significant vertical movement of the vortex ring over time. The ring in frame 145 is clearly lower in the tank than in frame 130, and the particles show reduced correlation, reflecting the ring's travel and surrounding fluid displacement.

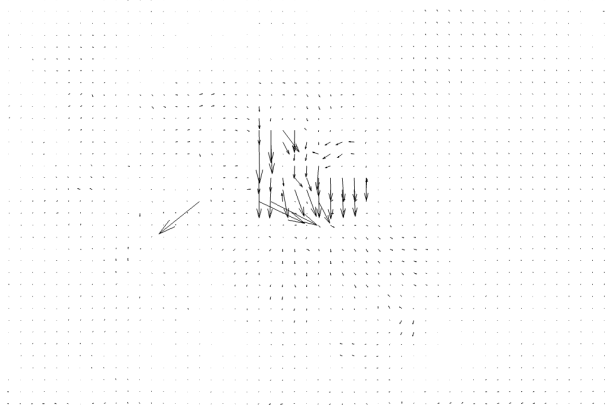
Frame = 70
Time = 2.212s



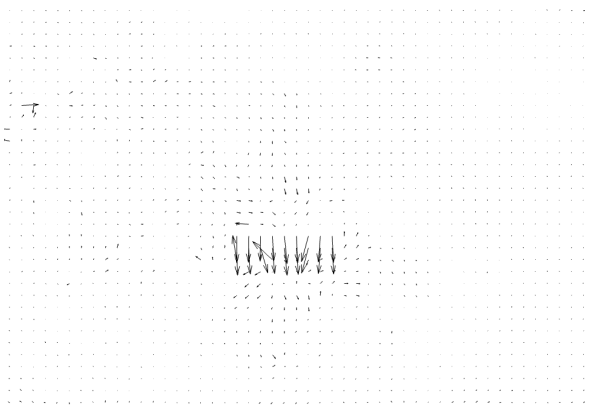
Frame = 85
Time = 2.677s



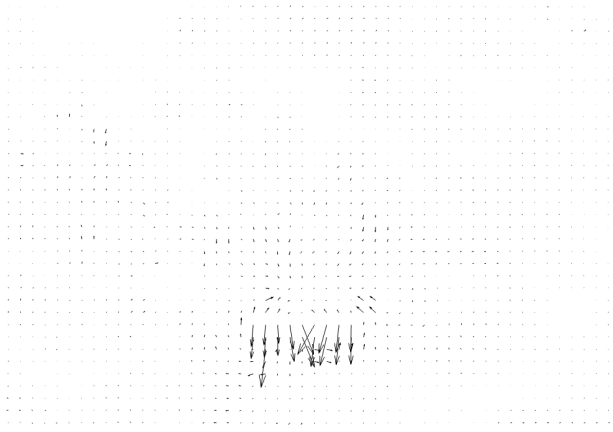
Frame = 100
Time = 3.224 s



Frame = 115
Time = 3.623 s



Frame = 130
Time = 4.118s



Frame = 145
Time = 4.545 s



Figure 4. Six PIV vector fields illustrating the temporal evolution of a single vortex ring, captured at 5-frame intervals between frame 70 (2.212 s) and frame 145 (4.545 s). In the first image (Frame 70, 2.212 s), a strong upward jet dominates. Nearly all arrows point vertically, indicating vigorous fluid ejection from the nozzle with little lateral motion. By Frame 85 (2.677 s), the leading edge of the forming ring begins to entrain surrounding fluid, as seen by inclined vectors at its periphery. Upward flow still dominates at the core, but angled arrows signal the inception of toroidal rotation. At Frame 100 (3.224 s), a coherent circular pattern emerges. Vectors near the center continue upward while those just outside curve inward and downward, evidencing a well-defined ring vortex in its developing stage. The fourth panel, Frame 115 (3.623 s), shows the vortex ring at near-peak circulation. Arrows trace a strong toroidal loop with minimal breakdown, and the core's upward velocity is balanced by inward flow feeding the ring structure. As the ring propagates in Frame 130 (4.118 s), its circulation begins to weaken. Vectors spread radially outward and decrease in length, indicating ring expansion and a slowdown of peak vertical velocity. Finally, in Frame 145 (4.545 s), the vortex ring enters its decay phase, with much smaller, more disordered arrows marking residual rotational motion and a dissipating toroidal core. (Scale bar: 10 cm/s.)

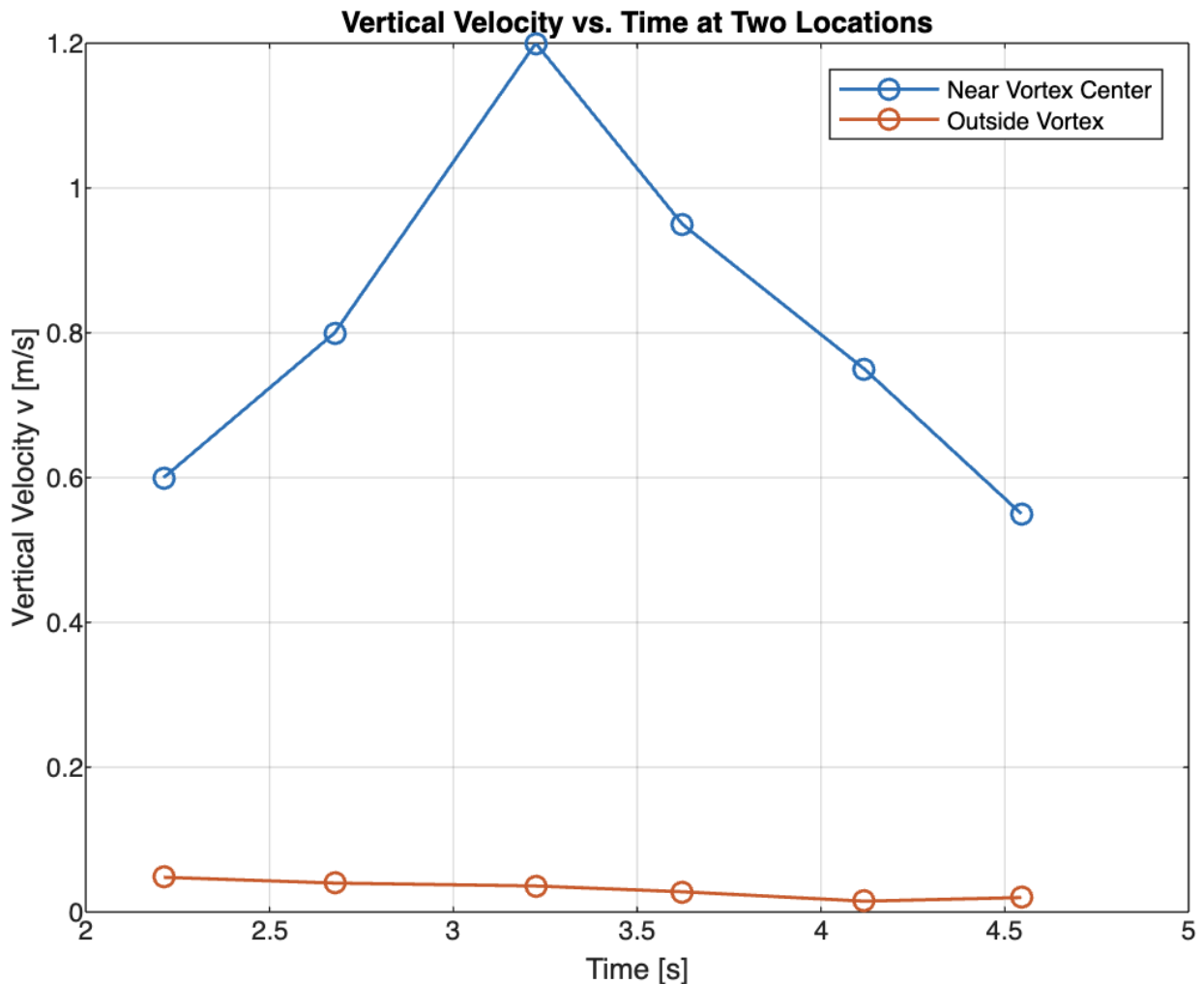


Figure 5:

This plot compares vertical velocity near the vortex center and at a location outside the vortex path. At ~2.2–3.2 seconds, velocity near the center spikes, matching the passage of the vortex. In contrast, the velocity outside the vortex remains low and flat, confirming that most flow is concentrated around the vortex core.

Discussion:

Using PIVLab, we captured how a vortex ring moves through a water tank over time. Our results showed clear phases: initial formation, growth with increasing circular velocity, and eventual decay. The strongest vertical motion occurred near the vortex center, while areas farther away remained mostly still. This supports the idea that vortex rings cause highly localized fluid motion.